

FRANKLINE MISANGO OYOLO

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SUMMARY

A Quantitative Developer utilizing C++ and Python with research experience in equities, futures, and options. I am seeking Quantitative roles in Foreign Exchange, Derivatives, and Investment Management, with an end-goal of innovating pricing, advisory, and client-centric solutions.

EDUCATION

The University of Hong Kong

Bachelor of Engineering in Artificial Intelligence— GPA: 3.73/4.3

Expected Dec 2027

Hong Kong SAR

ENTREPRENEURIAL AND WORK EXPERIENCE

Citi

Jun 2025 – Aug 2025

Markets Summer Analyst (Sales and Trading)

Hong Kong SAR

- Selected for a dual-rotation program across various trading desks, requiring rapid adaptation to diverse asset classes and market dynamics.
- Collaborated with global teams to integrate data into factor models, presenting actionable insights to regional senior management.
- Automated complex desk workflows using VBA, Python, and Excel to enhance efficiency and accuracy during high-volume trading sessions.
- Demonstrated resilience and technical agility by migrating real-time market data feeds to FPGA-based VHDL simulations for future systems.

The Swiss Federal Institute of Technology (EPFL)

Jul 2025 – Aug 2025

Doctoral Research Assistant

Lausanne, Switzerland

- Awarded the Summer @ EPFL Grant of CHF 5000 of a 0.8% competitive rate globally to Participate in the program within the Summer
- Engineered a multi-scale feature integration pipeline using Python to extract demographic and credit transactions at a 1000m radius/station
- Achieved 27% RMSE reduction over ST-GCN baselines with 13.4% gains from OSM integration; providing ablation studies for fiscal planners
- Published to prestigious ACM SIGSPATIAL, proposing a customized Machine Learning model for demand forecasting of fiscal transactions

BASF SE

Jan 2025 – Jun 2025

Machine Learning Engineer

Hong Kong SAR

- Built Python-based DiD and local-projection shock models generating 50k Monte Carlo paths, reducing DSO volatility by 42%
- Optimized global supplier networks via CVaR-constrained Monte Carlo simulation, cutting landed-cost variance to 25bps and \$2.3M in capital.
- Developed SHAP-interpretable valuation models decomposing enterprise drivers to guide semiconductor strategy toward a projected 114% EBIT.

Manulife Financial Corporation

Jun 2024 – Aug 2024

Applied Data Scientist

Hong Kong SAR

- Built metadata enrichment engine using Python; increasing actuarial tag coverage of 2.3M Azure Purview Assets from 4% to 65%
- Implemented column-level lineage tracking using Snowflake and GitOps, mapping premium data to IFRS 17 calculation layers
- Automated 300+ data quality checks with Great Expectations framework in C#, improving new business margin screening precision to 92%

Arithmax Research

Jan 2023 – Present

Principal Founder

Remote

- Writes and backtests diverse quantitative research problems, publishing papers and reports on trading strategies and market dynamics
- Designs ultra low-latency medium frequency trading infrastructure in C++ to achieve sub-millisecond event precision and high system uptime
- Builds high-throughput market-making infrastructure in Rust, C and C# to fasten advanced mathematical model with jump-diffusion dynamics
- Programs Icarus simulations for real VHDL/Verilog FPGAs with real-time market data feeds for future system migrations
- Integrates alternative data sources like satellite imagery, social sentiment, and weather into factor models, enhancing regime-shifts robustness

MARKETS APTITUDE

Cross-Regime Performance Analysis of an Algorithmic Strategy for a Diversified Leverage Index Fund Portfolio Caltech Investment Society

- Developed a risk-parity portfolio model using stochastic control equations to achieve 46.2 percent annualized returns.
- Mitigated volatility decay in leveraged ETFs via GARCH optimized weighting, reducing decay by 38 percent versus naive allocation.

Exploiting Arbitrage in Currency Crashes: The R-Zone Early Warning Strategy

HKU Trading Group

- Developed a logistic regression classifier with an 84 percent accuracy rate to identify FX regime shifts and predict currency crashes.
- Utilized FX-implied volatility surfaces to engineer hedging strategies, mitigating drawdown by 24 percent during market dislocations.

Intraday Momentum Breakout Strategy: A Volatility-Targeted Approach to E-mini Futures Trading

HKU Quantitative Society

- Developed a logistic regression classifier with an 84 percent accuracy rate to identify FX regime shifts and predict currency crashes.
- Utilized FX-implied volatility surfaces to engineer hedging strategies, mitigating drawdown by 24 percent during market dislocations.

PROGRAMMING PROJECTS

FeenQR: Automated Portfolio Analysis Application | ★ 142

C#, Python, AI

- Engineered an end-to-end automation platform for portfolio stress-testing and rebalancing across 40 distinct asset classes.
- Integrated real-time regime detection and neural network volatility forecasting to automate risk management for institutional-grade strategies.

TradeKernel: CUDA-Accelerated Market Making | ★ 310

C++, CUDA, High-Performance Computing

- Implemented GPU-accelerated finite-difference solvers for a 136x speedup, preserving 97.4 percent PnL efficiency during execution.
- Built an automated order flow toxicity tracker to reduce adverse selection by 41 percent through real-time imbalance-aware quoting.

SKILLS, INTERESTS & ACTIVITIES

Research: Options Pricing (Black-Scholes, Heston), Stochastic Calculus (HJB), Order Book Dynamics (LOB), PDE Solvers (Crank-Nicolson, FD)

Languages: English (Native), German (A2), French (A2), Mandarin (Beginner)

Programming: Python, C#, C++, Rust, CUDA (GPU Acceleration), VHDL/Verilog (FPGA)

Tools: MS Excel, VBA (Macro Automation), Bloomberg Terminal, Reuters Eikon

Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional, CFI Financial Modeling and Valuation Analyst (FMVA)

Extra Curricular: Chess (Blitz 1800, Rapid 1500), Leetcode (top 5%), Cycling, International travel

Self Readings : Financial Markets, Global Geopolitics, Chip Wars, Planetary Science, Evolutionary Biology, Zen Buddhism

Campus Involvement: HKU Trading Group TMT Software Developer, HKU Quantitative Researcher for Analysis of Algorithms

Awards: Amazon Web Services ABW Grant (2024, 2025), Python Software Foundation Grant (2024)